

Coupled Foldy-Lax T-Matrix: ($\Delta\rho$, $\Delta\lambda$, $\Delta\mu$) Resonance Extension

T. M. Nestor

Research School of Earth Sciences, Australian National University

1 Introduction

The Rayleigh-regime T-matrix for a cubic scatterer (see companion document *Cubic T-Matrix with the Full Elastodynamic Green's Tensor*) captures density, bulk modulus, and shear modulus contrasts through the self-consistent effective contrasts $\Delta\rho^*$, $\Delta\lambda^*$, and $\Delta\mu^*$. However, the displacement-only Foldy-Lax formulation that extends this to the resonance regime ($ka \sim 1$) captures only density scattering: the 3×3 sub-cell T-matrix $\mathbf{T}_{\text{sub}} = \omega^2 \Delta\rho^* V_{\text{sub}} \mathbf{I}_3$ has no mechanism to transmit stiffness contrasts between sub-cells, because stiffness scattering acts on the *strain* field, not the displacement field.

To include all three contrast types ($\Delta\rho$, $\Delta\lambda$, $\Delta\mu$) in the internal Foldy-Lax system, the state vector at each sub-cell must be enlarged from displacement alone (3 dof) to displacement plus Voigt strain (9 dof). The resulting $9N \times 9N$ coupled system—where $N = n^3$ is the number of sub-cells—is the subject of this document.

The coupled formulation:

1. Subdivides the cube of half-width a into n^3 sub-cells, each of half-width $a_{\text{sub}} = a/n$, in the Rayleigh regime.
2. Assigns a 9×9 local T-matrix to each sub-cell (density block + stiffness block).
3. Builds the 9×9 inter-sub-cell propagator from the Green's tensor G_{ij} , its first derivative $G_{ij,k}$, and its second derivative $G_{ij,kl}$.
4. Solves the $9N \times 9N$ Foldy-Lax system and extracts the 9×9 composite T-matrix.
5. Converts to the physical 6×6 displacement-traction T-matrix via the volume-integrated self-Green's tensor.

The implementation is activated by passing `coupled=True` to `compute_resonance_tmatrix()`.

2 Green's Tensor and Derivatives

2.1 Kupradze representation

The elastodynamic Green's tensor for a homogeneous isotropic full space satisfies (??) with the Kupradze solution

$$G_{ij}(\mathbf{x}) = P[\varphi(r) \delta_{ij} + \psi(r) \gamma_i \gamma_j], \quad P = \frac{1}{4\pi\rho\omega^2}, \quad (1)$$

where $r = |\mathbf{x}|$, $\gamma_i = x_i/r$, and

$$\mu \nabla^2 G_{ij} + (\lambda + \mu) \partial_i \partial_k G_{kj} + \rho \omega^2 G_{ij} = -\delta_{ij} \delta(\mathbf{x}). \quad (2)$$

The radial functions φ and ψ are sums of terms of the form $c \cdot e^{ikr} \cdot r^p$ with $p \in \{-1, -2, -3\}$:

$$\varphi(r) = k_S^2 \frac{e^{ik_S r}}{r} + ik_S \frac{e^{ik_S r}}{r^2} - \frac{e^{ik_S r}}{r^3} - ik_P \frac{e^{ik_P r}}{r^2} + \frac{e^{ik_P r}}{r^3}, \quad (3)$$

$$\psi(r) = -k_S^2 \frac{e^{ik_S r}}{r} - 3ik_S \frac{e^{ik_S r}}{r^2} + 3 \frac{e^{ik_S r}}{r^3} + k_P^2 \frac{e^{ik_P r}}{r} + 3ik_P \frac{e^{ik_P r}}{r^2} - 3 \frac{e^{ik_P r}}{r^3}. \quad (4)$$

2.2 Derivative rules

For each term $c \cdot e^{ikr} \cdot r^p$, the radial derivatives follow from the chain rule on composite exponential-power functions:

$$\frac{d}{dr}[e^{ikr} r^p] = e^{ikr} r^{p-1} (ikr + p), \quad (5)$$

$$\frac{d^2}{dr^2}[e^{ikr} r^p] = e^{ikr} r^{p-2} [(ikr)^2 + 2p(ikr) + p(p-1)]. \quad (6)$$

These are applied term-by-term to φ and ψ to obtain φ' , ψ' , φ'' , ψ'' .

2.3 First derivative $G_{ij,k}$

Using $\partial_k r = \gamma_k$ and $\partial_k \gamma_i = (\delta_{ik} - \gamma_i \gamma_k)/r$:

$$G_{ij,k} = P[c_1 \delta_{ij} \gamma_k + c_2 \gamma_i \gamma_j \gamma_k + c_3 (\delta_{ik} \gamma_j + \delta_{jk} \gamma_i)], \quad (7)$$

with three scalar coefficients:

$$c_1 = \varphi', \quad c_2 = \psi' - \frac{2\psi}{r}, \quad c_3 = \frac{\psi}{r}. \quad (8)$$

2.4 Second derivative $G_{ij,kl}$

The second spatial derivative involves seven independent tensor structures:

$$G_{ij,kl} = P \left[t_1 \delta_{ij} \delta_{kl} + t_2 \delta_{ij} \gamma_k \gamma_l + t_3 \gamma_i \gamma_j \delta_{kl} \right. \\ \left. + t_4 (\delta_{ik} \delta_{jl} + \delta_{jk} \delta_{il}) + t_5 (\delta_{il} \gamma_j \gamma_k + \delta_{jl} \gamma_i \gamma_k \right. \\ \left. + \delta_{ik} \gamma_j \gamma_l + \delta_{jk} \gamma_i \gamma_l) + t_7 \gamma_i \gamma_j \gamma_k \gamma_l \right], \quad (9)$$

with five independent scalar coefficients (note that t_3 appears in three of the seven structures):

Table 1. Scalar coefficients for the seven tensor structures in $G_{ij,kl}$.

Coeff.	Expression	Tensor structures
t_1	φ'/r	$\delta_{ij} \delta_{kl}$
t_2	$\varphi'' - \varphi'/r$	$\delta_{ij} \gamma_k \gamma_l$
t_3	$\psi'/r - 2\psi/r^2$	$\gamma_i \gamma_j \delta_{kl}$, 4 mixed terms
t_4	ψ/r^2	$\delta_{ik} \delta_{jl} + \delta_{jk} \delta_{il}$
t_7	$\psi'' - 5\psi'/r + 8\psi/r^2$	$\gamma_i \gamma_j \gamma_k \gamma_l$

3 Voigt Contraction

The coupled system requires contracting the Green's tensor derivatives into the 6-component Voigt strain space.

3.1 Voigt ordering

The Voigt ordering maps symmetric pairs of Cartesian indices to a single index $\alpha \in \{1, \dots, 6\}$:

$$\alpha \longleftrightarrow (p, q) : \quad 1 = (0, 0), \quad 2 = (1, 1), \quad 3 = (2, 2), \quad 4 = (1, 2), \quad 5 = (0, 2), \quad 6 = (0, 1), \quad (10)$$

corresponding to $(\varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx})$ in the (z, x, y) coordinate convention (axis labelling $1 \rightarrow z, 2 \rightarrow x, 3 \rightarrow y$).

3.2 Voigt propagator blocks

From $G_{ij,k}$ and $G_{ij,kl}$, three Voigt-contracted matrices are formed:

$$C_{i\alpha} = \begin{cases} G_{ip,p} & \text{if } \alpha \leftrightarrow (p, p), \\ G_{ip,q} + G_{iq,p} & \text{if } \alpha \leftrightarrow (p, q), p \neq q, \end{cases} \quad (11)$$

$$H_{\alpha j} = \begin{cases} G_{pj,p} & \text{if } \alpha \leftrightarrow (p, p), \\ G_{pj,q} + G_{qj,p} & \text{if } \alpha \leftrightarrow (p, q), p \neq q, \end{cases} \quad (12)$$

$$S_{\alpha\beta} = \text{double Voigt contraction of } G_{ij,kl}, \quad (13)$$

with dimensions $\mathbf{C} \in \mathbb{C}^{3 \times 6}$, $\mathbf{H} \in \mathbb{C}^{6 \times 3}$, $\mathbf{S} \in \mathbb{C}^{6 \times 6}$.

The block $\mathbf{C}$ maps a Voigt stress source to a displacement response; $\mathbf{H}$ maps a force source to a Voigt strain response; $\mathbf{S}$ maps a Voigt stress source to a Voigt strain response.

Reciprocity. By the symmetry $G_{ij,k} = G_{ji,k}$ of the Green's tensor, the relationship $\mathbf{H} = \mathbf{C}^\top$ holds. This is verified numerically in the implementation.

4 The 9×9 Coupled System

4.1 State vector

At each sub-cell n , the state vector combines displacement and Voigt strain:

$$\boldsymbol{\psi}_n = (u_1, u_2, u_3, \varepsilon_{V1}, \dots, \varepsilon_{V6})_n \in \mathbb{C}^9. \quad (14)$$

4.2 Local T-matrix (9×9)

Each sub-cell has a block-diagonal local T-matrix:

$$\mathbf{T}_{\text{loc}} = \begin{pmatrix} \omega^2 \Delta \rho^* V_{\text{sub}} \mathbf{I}_3 & \mathbf{0}_{3 \times 6} \\ \mathbf{0}_{6 \times 3} & V_{\text{sub}} \Delta \mathbf{c}_V^* \end{pmatrix}, \quad (15)$$

where $V_{\text{sub}} = (2a_{\text{sub}})^3$ is the sub-cell volume, $\Delta \rho^* = \Delta \rho \cdot A_u$ is the self-consistent effective density contrast, and $\Delta \mathbf{c}_V^*$ is the 6×6 effective stiffness contrast in Voigt form:

$$\Delta \mathbf{c}_V^* = \begin{pmatrix} \mathbf{D}^* & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{S}^* \end{pmatrix}, \quad D_{ij}^* = \begin{cases} \Delta \lambda^* + 2\Delta \mu_{\text{diag}}^* & i = j \\ \Delta \lambda^* & i \neq j \end{cases}, \quad S_{ij}^* = 2\Delta \mu_{\text{off}}^* \delta_{ij}. \quad (16)$$

The density block produces a scattered force proportional to displacement; the stiffness block produces a scattered stress perturbation proportional to strain. Both use the self-consistent

effective contrasts from the Rayleigh T-matrix computation (`effective_contrasts.py`).

4.3 Inter-sub-cell propagator

The 9×9 propagator between sub-cells m and n ($m \neq n$) is:

$$\mathbf{P}_{mn} = \left(\begin{array}{cc} G_{ij} & C_{i\alpha} \\ H_{\alpha j} & S_{\alpha\beta} \end{array} \right) \Big|_{\mathbf{x}=\mathbf{x}_m-\mathbf{x}_n}, \quad (17)$$

evaluated at the separation vector $\mathbf{x}_m - \mathbf{x}_n$. The diagonal blocks ($m = n$) are zero—the self-interaction is absorbed into $\mathbf{T}_{\text{loc}}$ via the Rayleigh self-consistent amplification factors.

4.4 $9N \times 9N$ Foldy-Lax equation

Assembling the block-diagonal local T-matrix $\tilde{\mathbf{T}} = \mathbf{I}_N \otimes \mathbf{T}_{\text{loc}}$ and the off-diagonal propagator $\tilde{\mathbf{P}}$, the coupled Foldy-Lax system reads:

$$(\mathbf{I}_{9N} - \tilde{\mathbf{P}} \tilde{\mathbf{T}}) \boldsymbol{\psi}_{\text{exc}} = \boldsymbol{\psi}_{\text{inc}}, \quad (18)$$

where $\boldsymbol{\psi}_{\text{inc}}$ is the incident field and $\boldsymbol{\psi}_{\text{exc}}$ is the exciting (self-consistent) field at all sub-cells.

4.5 Incident field construction

To extract the full 9×9 composite T-matrix, nine independent incident patterns are needed:

- **Columns 0–2** (displacement inputs): uniform unit displacement $u_i^{\text{inc}}(\mathbf{x}_m) = \hat{e}_p$, zero incident strain $\boldsymbol{\varepsilon}_V^{\text{inc}} = \mathbf{0}$.
- **Columns 3–8** (strain inputs): linear displacement variation $u_i^{\text{inc}}(\mathbf{x}_m) = \varepsilon_{ij}^0(x_{m,j} - \bar{x}_j)$ plus uniform Voigt strain $\boldsymbol{\varepsilon}_V^{\text{inc}} = \hat{e}_\alpha$.

A phase factor $e^{i\mathbf{k} \cdot \mathbf{x}_m}$ is applied to all patterns for resonance-regime accuracy.

4.6 Composite T-matrix extraction

The 9×9 composite T-matrix is obtained by summing the scattered contributions from all sub-cells:

$$\mathbf{T}_{\text{comp}}[:, q] = \sum_{n=1}^N \mathbf{T}_{\text{loc}} \cdot \boldsymbol{\psi}_{\text{exc}}[9n:9(n+1), q], \quad (19)$$

where $q \in \{1, \dots, 9\}$ indexes the nine independent incident patterns. The result $\mathbf{T}_{\text{comp}}$ maps incident $(u_i, \varepsilon_{V\alpha})$ to scattered $(F_i, \Delta\sigma_{V\alpha}^*)$ —force monopole and stress dipole amplitudes.

5 Self-Interaction and Volume Integrals

The self-interaction (diagonal blocks) of the Foldy-Lax propagator requires the volume integrals of G_{ij} , $G_{ij,k}$, and $G_{ij,kl}$ over the cube. These are already computed for the Rayleigh T-matrix; here we collect the relevant results for the output conversion stage.

5.1 First-derivative integral

By the symmetry of the cube $([-a, a]^3)$ and the odd parity of $G_{ij,k}$ under inversion $\mathbf{x} \rightarrow -\mathbf{x}$:

$$\int_{\text{cube}} G_{ij,k} dV = 0. \quad (20)$$

This means the displacement-strain and strain-displacement cross-coupling vanishes in the self-Green's tensor: $\mathbf{C}_{\text{self}} = \mathbf{0}$, $\mathbf{H}_{\text{self}} = \mathbf{0}$.

5.2 Second-derivative integral

The volume integral of $G_{ij,kl}$ decomposes under cubic symmetry exactly as the integral tensor from the companion document:

$$\int_{\text{cube}} G_{ij,kl} dV = A \delta_{ij} \delta_{kl} + B(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + C E_{ijkl}, \quad (21)$$

where $E_{ijkl} = \delta_{ij} \delta_{kl} \delta_{ik}$ is the cubic tensor (nonzero only when $i = j = k = l$).

Each coefficient receives a static (Eshelby) and smooth (radiation) contribution:

$$A = A_{\text{stat}} + A_{\text{smooth}}, \quad B = B_{\text{stat}} + B_{\text{smooth}}, \quad C = C_{\text{stat}} + C_{\text{smooth}}. \quad (22)$$

Static Eshelby coefficients

The static contributions arise from the principal-value volume integral of the Kelvin Green's tensor, reduced to surface integrals over the cube faces via the divergence theorem:

$$A_{\text{stat}} = \frac{(\sqrt{3} - 2\pi) \alpha^2 - \sqrt{3} \beta^2}{6\pi \rho \alpha^2 \beta^2}, \quad (23)$$

$$B_{\text{stat}} = \frac{\alpha^2 - \beta^2}{2\sqrt{3} \pi \rho \alpha^2 \beta^2}, \quad (24)$$

$$C_{\text{stat}} = \frac{(-5\sqrt{3} + 2\pi)(\alpha^2 - \beta^2)}{6\pi \rho \alpha^2 \beta^2}. \quad (25)$$

Note that $C_{\text{stat}}/B_{\text{stat}} = -5 + 2\pi/\sqrt{3} \approx -1.372$ is a pure geometric constant (shape-dependent, material-independent). Both B_{stat} and C_{stat} are proportional to $(\alpha^2 - \beta^2)$ and vanish in the mathematical limit $\alpha \rightarrow \beta$, where the T-matrix becomes isotropic (sphere-like). For physical

materials $\alpha > \beta$ always holds (stability requires $K > 0$, so $\alpha/\beta = \sqrt{(\lambda + 2\mu)/\mu} > \sqrt{4/3}$).

Smooth radiation corrections

The smooth (radiation) contributions come from integrating the regular part of the Green's tensor and its derivatives over the cube. At leading order in ka :

$$A_{\text{smooth}} \approx -\frac{2i a^3 \omega^3}{3\pi \beta^5 \rho} + \mathcal{O}(a^5 \omega^5), \quad (26)$$

$$B_{\text{smooth}} \approx \frac{i a^3 \omega^3 (\alpha^5 - \beta^5)}{3\pi \alpha^5 \beta^5 \rho} + \mathcal{O}(a^5 \omega^5), \quad (27)$$

$$C_{\text{smooth}} \approx \frac{2i a^7 \omega^7 (\beta^9 - \alpha^9)}{4725\pi \alpha^9 \beta^9 \rho} + \mathcal{O}(a^9 \omega^9). \quad (28)$$

Since C_{smooth} first appears at $\mathcal{O}((ka)^7)$ while C_{stat} is frequency-independent, the cubic anisotropy is dominated by the static Eshelby depolarization in the Rayleigh regime.

5.3 Voigt form of the self-integral

Contracting (??) into Voigt indices gives the 6×6 matrix $\mathbf{S}_V$:

$$\mathbf{S}_V = \begin{pmatrix} \mathbf{D}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_S \end{pmatrix}, \quad (29)$$

with D-block entries

$$(D_S)_{ii} = A + 2B + C, \quad (D_S)_{ij} = B \quad (i \neq j), \quad (30)$$

and S-block entries

$$(S_S)_{ii} = 2(A + B), \quad (S_S)_{ij} = 0 \quad (i \neq j). \quad (31)$$

The block-diagonal structure of $\mathbf{S}_V$ (vanishing D–S coupling) is a consequence of the cubic symmetry of the cube and the decomposition (??) involving only the three independent tensors $\delta_{ij}\delta_{kl}$, $\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$, and E_{ijkl} .

6 Near-Field Exact 6×6 T-Matrix

The composite T-matrix $\mathbf{T}_{\text{comp}}$ operates in the 9-component $(u_i, \varepsilon_{V\alpha})$ space, but the physical observable is the 6×6 T-matrix in the displacement-traction basis $(u_z, u_x, u_y, t_{zz}, t_{xz}, t_{yz})$. Converting between these requires careful treatment of the self-interaction Green's tensor.

6.1 Problem: far-field traction approximation

For the displacement-only (3×3) formulation, the scattered traction is approximated using the far-field radiation pattern $(t_i \approx ikc_{ijkl}G_{kl} \dots)$. This fails for densely packed sub-cells where

near-field interactions dominate, and it completely misses the stiffness scattering contribution to the traction.

6.2 Three-stage pipeline

The near-field exact 6×6 T-matrix is constructed via a three-stage pipeline:

$$\boxed{\mathbf{T}_{6 \times 6} = \mathbf{G}_{\text{self}} \cdot \mathbf{T}_{\text{comp}} \cdot \mathbf{M}_{\text{in}}}, \quad (32)$$

where:

Stage 1: Input conversion $\mathbf{M}_{\text{in}}$ (9×6). Maps the incident displacement-traction vector $(u_z, u_x, u_y, t_{zz}, t_{xz}, t_{yz})$ to the 9-component state $(u_i, \varepsilon_{V\alpha})$:

$$\mathbf{M}_{\text{in}} = \begin{pmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{S}(k_x, k_y) \end{pmatrix}, \quad (33)$$

where $\mathbf{S}(k_x, k_y)$ is the 6×6 strain extraction matrix that maps $(u_z, u_x, u_y, t_{zz}, t_{xz}, t_{yz})$ to Voigt strain $(\varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx})$. The strain extraction follows from the constitutive relation on a $z = \text{const}$ surface:

$$\begin{aligned} \varepsilon_{zz} &= \frac{t_{zz} - \lambda i k_x u_x - \lambda i k_y u_y}{\lambda + 2\mu}, & \varepsilon_{xx} &= i k_x u_x, \\ \varepsilon_{yy} &= i k_y u_y, & 2\varepsilon_{xy} &= i k_y u_x + i k_x u_y, \\ 2\varepsilon_{zy} &= t_{yz}/\mu, & 2\varepsilon_{zx} &= t_{xz}/\mu. \end{aligned} \quad (34)$$

Stage 2: Composite T-matrix $\mathbf{T}_{\text{comp}}$ (9×9). The coupled Foldy-Lax output, mapping incident $(u_i, \varepsilon_{V\alpha})$ to scattered $(F_i, \Delta\sigma_{V\alpha}^*)$.

Stage 3: Output conversion $\mathbf{G}_{\text{self}}$ (6×9). Converts the scattered source amplitudes $(F_i, \Delta\sigma_{V\alpha}^*)$ to the scattered displacement-traction $(u_i, t_i)^{\text{scat}}$ via the volume-integrated self-Green's tensor:

$$\mathbf{G}_{\text{self}} = \begin{pmatrix} \Gamma_0 \mathbf{I}_3 & \mathbf{0}_{3 \times 6} \\ \mathbf{0}_{3 \times 3} & \mathbf{C}_{\text{trac}} \cdot \mathbf{S}_V \end{pmatrix}, \quad (35)$$

where $\Gamma_0 = \int_{\text{cube}} G_{ii} dV/3$ is the scalar volume integral of the Green's tensor, and $\mathbf{S}_V$ is the Voigt form of the self-integral (??).

The block-diagonal structure follows from the vanishing of the cross-coupling integrals ($\mathbf{C}_{\text{self}} = \mathbf{H}_{\text{self}} = \mathbf{0}$, §??).

6.3 Traction extraction matrix

The 3×6 matrix $\mathbf{C}_{\text{trac}}$ extracts the traction on a $z = \text{const}$ surface from the Voigt stress:

$$\mathbf{C}_{\text{trac}} = \begin{pmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \\ 0 & 0 & 0 & 0 & \mu & 0 \end{pmatrix}, \quad (36)$$

mapping $(\sigma_{zz}, \sigma_{xx}, \sigma_{yy}, \sigma_{xy}, \sigma_{zy}, \sigma_{zx})_V$ to (t_{zz}, t_{xz}, t_{yz}) .

6.4 Complete formula

Combining all three stages:

$$\mathbf{T}_{6 \times 6}(k_x, k_y) = \begin{pmatrix} \Gamma_0 \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{\text{trac}} \mathbf{S}_V \end{pmatrix} \cdot \mathbf{T}_{\text{comp}} \cdot \begin{pmatrix} \mathbf{I}_3 & \mathbf{0} \\ \mathbf{S}(k_x, k_y) \end{pmatrix}. \quad (37)$$

This is the near-field exact result, valid at all frequencies and for all contrast types. The dependence on horizontal wavenumbers (k_x, k_y) enters only through the strain extraction matrix $\mathbf{S}$. Figure ?? summarises the full data flow, from Kennett partial-wave amplitudes, through the 9-component composite state of the scatterer, and back.

7 Implementation Notes

7.1 Function map

Table ?? maps the key functions in the implementation to the equations in this document.

Table 2. Function-to-equation reference for the coupled Foldy-Lax implementation.

Function	Equation	Size
<code>elastodynamic_greens_deriv</code>	(??), (??), (??)	$3^2, 3^3, 3^4$
<code>_voigt_contract</code>	(??)–(??)	$3 \times 6, 6 \times 3, 6 \times 6$
<code>_propagator_block_9x9</code>	(??)	9×9
<code>_sub_cell_tmatrix_9x9</code>	(??)	9×9
<code>_build_incident_field_coupled</code>	§??, §4.5	$9N \times 9$
<code>_solve_coupled</code>	(??), (??)	$9N \times 9N$
<code>_volume_integral_voigt</code>	(??)–(??)	6×6
<code>_traction_from_voigt_strain</code>	(??)	3×6
<code>_voigt_tmatrix_near_field</code>	(??)	6×6
<code>strain_from_displacement_traction</code>	(??)	6×6
<code>effective_stiffness_voigt</code>	(??)	6×6

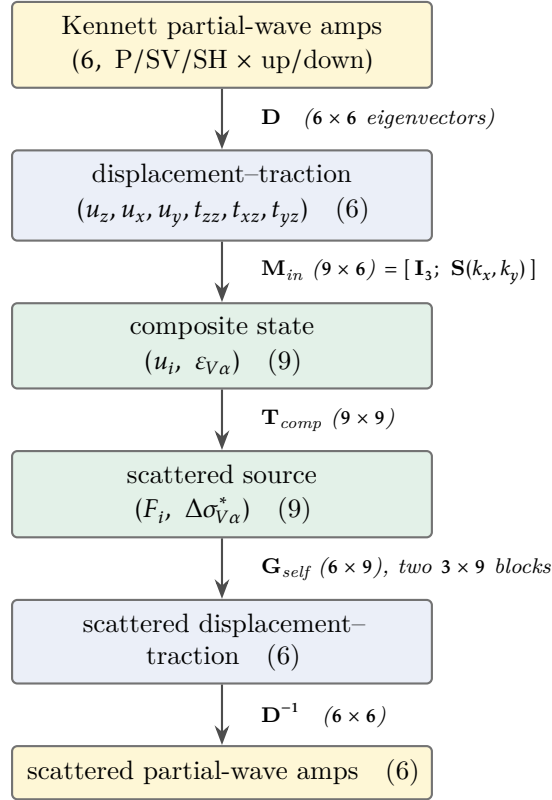


Figure 1. Data flow of the three-stage pipeline (??). The 6-component displacement–traction vector $(u_z, u_x, u_y, t_{zz}, t_{xz}, t_{yz})$ is the bridge between the Kennett partial-wave basis (decomposed by the 6 × 6 eigenvector matrix $\mathbf{D}$, two 3 × 6 blocks) and the 9-component composite state $(u_i, \varepsilon_{V\alpha})$ of the scatterer. The three in-plane Voigt stresses dropped on the Kennett side are restored by the strain-extraction matrix $\mathbf{S}$ inside $\mathbf{M}_{in}$.

7.2 Solver options

Two solvers are available:

- **Direct (LU):** `neumann_order = 0` (default). Exact solution via `np.linalg.solve`. Cost: $\mathcal{O}((9N)^3)$.
- **Truncated Neumann series:** `neumann_order > 0`. Iterative: $\psi_{exc} = \sum_{k=0}^K (\tilde{\mathbf{P}}\tilde{\mathbf{T}})^k \psi_{inc}$. Cost per iteration: $\mathcal{O}((9N)^2)$; converges when $\|\tilde{\mathbf{P}}\tilde{\mathbf{T}}\| < 1$.

7.3 Computational cost

The coupled system has size $9N \times 9N$ compared to $3N \times 3N$ for the displacement-only formulation. The direct solve cost scales as $(9N)^3/(3N)^3 = 27$ times larger. Propagator assembly scales as $9^2/3^2 = 9$ times larger per sub-cell pair, with the additional cost of computing $G_{ij,k}$ and $G_{ij,kl}$ (five radial functions vs. two).

For moderate N (e.g. $n = 3$, $N = 27$, system size 243), the direct solve remains tractable. For larger N , the Neumann series or iterative Krylov solvers are recommended.

7.4 Backward compatibility

Setting `coupled=False` (default) recovers the displacement-only $3N \times 3N$ system. When $\Delta\lambda = \Delta\mu = 0$ (density-only contrast), the coupled system produces the same 3×3 displacement T-matrix as the uncoupled system, since the stiffness block of $\mathbf{T}_{\text{loc}}$ vanishes and the strain channels decouple.

7.5 Validation

Two key consistency checks are built into the test suite:

1. **Rayleigh limit** ($n = 1$, **coupled**): With a single sub-cell ($N = 1$), no inter-sub-cell coupling exists and the composite T-matrix must equal the single-cell Rayleigh result.
2. **Density-only limit** ($\Delta\lambda = \Delta\mu = 0$, **coupled**): With zero stiffness contrast, the coupled 9×9 system must recover the displacement-only 3×3 T-matrix identically.